

Name: M. Tasca Razzaq
Roll No: 22C-7618
Section: 5A
Assignment: 1.

QUESTION No: 1

Solution:-

1. AM broadcast Radio's:

535 KHz - 1605 KHz.

$$\lambda_{\min} = \frac{c}{f_{\min}} = \frac{3 \times 10^8}{535 \times 10^3} = 560.75 \text{ m.}$$

$$\lambda_{\max} = \frac{c}{f_{\max}} = \frac{3 \times 10^8}{1605 \times 10^3} = 186.91 \text{ m.}$$

560.75 m to 186.91 m. Wavelength Range.

2. FM broadcast Radio's:

88 MHz to 108 MHz.

$$\lambda_{\min} = \frac{c}{f_{\min}} = \frac{3 \times 10^8}{88 \times 10^6} = 3.409 \text{ m.}$$

$$\lambda_{\max} = \frac{3 \times 10^8}{108 \times 10^6}$$

$$= 2.78 \text{ m.}$$

3.409 m to 2.78 m.

Wavelength Range.

3. VHF Civil Aviation Band:

108 MHz to 136 MHz.

2.778 m to 2.206 m. (Given)

4. GSM Cellular (Uplink):

890 MHz to 915 MHz.

$$\lambda_{\min} = \frac{3 \times 10^8}{88 \times 10^6}$$

$$= 0.337 \text{ m.}$$

$$\lambda_{\max} = \frac{3 \times 10^8}{915 \times 10^6}$$

$$= 0.327 \text{ m.}$$

0.337 m to 0.327 m. Wavelength Range.

5. GSM Cellular (Downlink):

925 MHz to 960 MHz.

$$\lambda_{\min} = \frac{3 \times 10^8}{925 \times 10^6}$$

$$= 0.324 \text{ m}$$

$$\lambda_{\max} = \frac{3 \times 10^8}{960 \times 10^6}$$

$$= 0.3125 \text{ m}$$

0.324 m to 0.3125 m. Wavelength Range.

6. Wi-Fi 802.11 b/g/n.

2.400 GHz to 2.497 GHz.

$$\lambda_{\min} = \frac{3 \times 10^8}{2.400 \times 10^9}$$

$$= 0.125 \text{ m}$$

$$\lambda_{\max} = \frac{3 \times 10^8}{2.497 \times 10^9}$$

$$= 0.120 \text{ m.}$$

0.125 m to 0.120 m. Wavelength Range

7. Wi-Fi 802.11 ac

4.915 GHz to 5.825 GHz.

$$\lambda_{\min} = \frac{3 \times 10^8}{4.915 \times 10^9}$$

$$= 0.0610 \text{ m}$$

$$\lambda_{\max} = \frac{3 \times 10^8}{5.825 \times 10^9}$$

$$= 0.0515 \text{ m}$$

0.0610 m to 0.0515 m Wavelength Range

8. X-band Police Radar:-

$$10.525 \text{ GHz}$$

$$\lambda = \frac{3 \times 10^8}{10.525 \times 10^9}$$

$$= 0.0285 \text{ m}$$

$$0.0285 \text{ m}$$

Wavelength Range

9. Ka-band Photo-Radar:-

$$34.2 \text{ GHz to } 34.4 \text{ GHz}$$

$$\lambda_{\min} = \frac{3 \times 10^8}{34.2 \times 10^9}$$

$$= 0.00877 \text{ m}$$

$$\lambda_{\max} = \frac{3 \times 10^8}{34.4 \times 10^9} = 0.00872 \text{ m}$$

0.00879 m to 0.00872 m Wavelength range

10. Wi-Fi 802.11 ad.

$$\lambda_{\min} = \frac{3 \times 10^8}{57.05 \times 10^9} = 0.00525 \text{ m}$$

$$\lambda_{\max} = \frac{3 \times 10^8}{71.0 \times 10^9} = 0.00422 \text{ m}$$

$$0.00525 \text{ m to } 0.00422 \text{ m}$$

11. Automotive Long Range Radar:

$$76 \text{ GHz to } 77 \text{ GHz}$$

$$\lambda_{\min} = \frac{3 \times 10^8}{76 \times 10^9} = 0.00394 \text{ m}$$

$$\lambda_{\max} = \frac{3 \times 10^8}{77 \times 10^9} = 0.00389 \text{ m}$$

$$0.00394 \text{ m to } 0.00389 \text{ Wavelength}$$

QUESTION No 2:-

Frequency	Loss in dB/100 m	Loss in dB/100 feet.
100 MHz	6.9	2.1
500 MHz	16.1	5.1
2500 MHz	46.5	14.2

Attenuation in dB is given

$$A = -10 \log \left(\frac{P_o}{P_i} \right)$$

P_o is output power.

P_i is input power.

$$\Rightarrow -\frac{A}{10} = \log \frac{P_o}{P_i}$$

$$P_o = P_i (10^{-A/10})$$

Now length of cable = 25 feet = 7.62m

A) $F = 100 \text{ MHz}$

$$\frac{A}{100 \text{ m}} = -6.9 \text{ dB} \quad \{ \text{Attenuation per 100m} \}$$

let C_n be given

$$C_n / 100 \text{ m} = -6.9 \text{ dB}$$

$$C_n / 7.62 \text{ m} = \frac{-6.9 \times 100}{7.62} = -0.526$$

$$\Rightarrow A / 7.62 \text{ m} = +0.526 \text{ dB} \quad \{ \text{Attenuation for 7.62 m or 25 feet} \}$$

$$\therefore P_o = 10 \text{ mW} \left(10^{\frac{-0.526}{10}} \right) \\ = 8.859 \text{ mW}$$

$$b) f = 500 \text{ Hz}$$

$$A / 100 \text{ m} = 166 \text{ dB}$$

$$A / 1.62 \text{ m} = 10237 \text{ dB}$$

$$P_o = 10 \text{ m} \left(10^{\frac{-10237}{10}} \right) \\ = 7.459 \text{ mW}$$

$$c) f = 23204 \text{ Hz}$$

$$\frac{A}{100 \text{ m}} = 46.5 \text{ dB}$$

$$A / 1.62 \text{ m} = 3.54$$

$$7.62$$

$$\Rightarrow P_o = 10 \text{ m} \left(10^{\frac{-3.54}{10}} \right) \\ = 4.426 \text{ mW}$$

QUESTION No: 3.

$$A). \text{ Signal-to-noise ratio} = 10 \log_{10} \frac{S}{N}$$

So putting values

$$30 \text{ dB} = 10 \log_{10} \frac{S}{N}$$

$$\log_{10} \frac{S}{N} = \frac{30}{10}$$

$$\frac{S}{N} = 10^{\frac{30}{10}}$$

$$\frac{S}{N} = 1000.$$

and equation for channel capacity in bit/s is

$$C = B \log_2 \left(1 + \frac{S}{N} \right).$$

Substitute 36,000 bps for C and 100 for $\frac{S}{N}$.
So,

$$36,000 = B \log_2 (1 + 1000)$$

$$B = \frac{36000}{\log_2 (1001)} \text{ Hz}$$

$$= \frac{36000}{9.965784284662087} \text{ Hz}.$$

So

$B = 3.6118 \text{ KHz}$ is required bandwidth.

B) Channel capacity.

$$C = B \log_2 \left(1 + \frac{S}{N} \right).$$

Substitute 25 kbps for C and 500 for $\frac{S}{N}$.

$$25 \text{ Kbps} = B \log_2 (1+500)$$

$$B = \frac{25}{\log_2 (501)} \text{ KHz.}$$

$$= \frac{25 \times 10^3}{\log_2 (501)} \text{ Hz.}$$

$$B = 2787.48 \text{ Hz.}$$

So

$B = 2.787 \text{ KHz}$ is required bandwidth.

QUESTION No: 4.

A) Channel Response:

It requires an ideal amplitude response

$$|H(\omega)| = 1.$$

and non ideal phase is given

$$\angle H(\omega) = -\omega t_0 - k \sin(\sin(\omega T)), \quad k \ll 1.$$

Using Fourier transform and Frequency Domain Analysis

Output $Y(\omega)$ is

$$Y(\omega) = G(\omega) \cdot H(\omega)$$

So

$$H(\omega) = e^{-j\omega t_0} \cdot e^{jk \sin(\sin(\omega T))}.$$

$$\therefore H(\omega) = e^{j\phi(\omega)}$$

where

$$e^{jk \sin(\sin(\omega T))}$$

$$\approx 1 - jk \sin(\sin(\omega T)).$$

$$H(\omega) \approx e^{-j\omega t_0} (1 - jk \sin(\sin(\omega T))).$$

and output signal in time domain is

$$y(t) = g(t - t_0) - jk \int_{-\infty}^{\infty} G(\omega) e^{j\omega t} \sin(\sin(\omega T)) d\omega$$

So output is a delayed version of input pulse $g(t - t_0)$ due to non ideal phase response.

B) Effect on TDM:

1. Impact:

- TDM involves multiple signal sharing a single communication channel by dividing time into slots.
- Non ideal response cause ISI due to distortion.
- It can cause distortion and overlapping between consecutive time slots which reduce integrity of individual signal.
- Sinusoidal phase $\propto \sin(\sin(\omega T))$, cause timing jitter, affecting synchronization between signals.

2. To Minimise (Mitigate) Impact:

- Employ equalisers to compensate for phase distortion.
- Implement forward error correction codes to recover from errors caused by ISI.
- Use adaptive filters to dynamically adjust to phase distortion over varying frequencies.

Effect on FDM:

- FDM involves multiple signals being transmitted simultaneously at different frequencies.
- Non ideal phase affect each phase to crosstalk between channels.
- The distortion varies over frequency, resulting in phase shift that cause frequency to interfere with adjacent channels.
- The harmonic distortion can overlap with other frequency bands.

2) To Mitigate Impact:

- Use guard bands between frequency channels to reduce interference.
- Implement phase correction filters to compensate for phase shift introduced by channel

QUESTION No 5:

- It include filter's transfer function, its realizability and how to make it realizable is
- first, necessary to find filter impulse response.
 - then we checking whether impulse response is causal or not.
 - last we will consider how to make it almost feasible

So determining impulse response $h(t)$.

$$H(\omega) = \frac{4(10^5)}{\omega^2 + 10^{10} + j\omega 10^5}$$

Inverse transform is

$$h(t) = 2(10^3) \times e^{-10^5 |t - t_0|}$$

$h(t)$ is non zero for all t . $t < 0$;
reactive and non causal making it unrealizable.

We require $h(t)$ less than to be 1%. when $t < 0$.
Maximum value of

$$h(t) \text{ at } t = t_0$$
$$h(t_0) = 2(10^3)$$
$$= 2000$$

We want $h(t) < 1\%$ of 2000

$$\text{So } h(t) < 20$$

So we finding t_0 .

$$h(t) = 2(10^3) \times e^{-10^5 t_0} < 20$$
$$e^{-10^5 t_0} = 1$$

$$= t_0 > -\ln\left(\frac{1}{100}\right)$$
$$10^5$$

$$= t_0 > 4.605 \times 10^{-5} \text{ seconds.}$$

$$t_0 = 4.605 \times 10^{-4} \text{ seconds.} = 0.4605 \text{ milliseconds.}$$

Let's choose t_0 to be ten times this value